MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 7782

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen, ceramic-metal

See figure 1

Mounting position: Any

Weight: 4.8 ounces (136 grams) nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	Ef Vac	epy kv	epx kv	Ebb gVdc	egy" v	egx v	Ib a	Ecc Vdc	Ip A
Maximum:	6.6	12.0 (Note 1)	12.0 (Note 2)	---	600 (Note 3)	200	350	150	5.0
Minimum:	5.8	---	5% epy	0.3	175	---	---	---	---
TEST CONDITIONS:	6.3	12.0	---	---	150	---	---	---	---

ABSOLUTE RATINGS:

Parameter: / Unit:	Ib Adc	prr ---	Eres Vac	Pb --	tk sec	dik/dt a/µs	TA °C	tJ µs (Note 7)	Cooling ---
Maximum:	0.2	Note 5	6.6	4.0x10⁹ (Note 8)	---	2,000	+150	0.005	Note 6
Minimum	---	---	5.8	---	180	---	-55	---	---
TEST CONDITIONS:	---	1,500	6.3	---	180	---	---	---	---

GENERAL:

Qualification - Required (see note 4)

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3241	Heater current (cathode)	-		0.65	II	If	3.5	7.0	Aac
3241	Heater current (reservoir)	-		0.65	II	Ires	1.0	2.5	Aac
3267	Instantaneous starting	9, 10	epy = 12 kv (min); Ef = Eres = 6.8 Vac	0.65	II	---	---	---	---
(A) 3247	DC anode voltage for conduction	9, 11	Ef = Eres = 5.8 Vac	0.65	II	Ebb	---	300	Vdc
(A) 3246	Operation (1)	9, 12	epy = 14.0 kv; Ef = Eres = 5.8 Vac	0.65	II	egy	---	150	v
(A) 3246	Operation (1A)	-	Operation (1); Ef = Eres = 6.8 Vac	0.65	II	egy	---	150	v
3251	Pulse emission	-	ik = 350 a; tp = 5.0 µs ± 10%; prr = 60 ± 10%; tr = 0.5 µs (max); specified time interval = 2.5 µs	0.65	II	egk	---	200	v
	Quality conformance inspection, part 2								
3256	Anode delay time	-	Operation (1); t = 120 seconds	---	---	tad	---	0.5	µs
3256	Anode delay time drift	13	Anode delay time	---	---	Δtad	---	0.10	µs
3261	Time jitter	14	Operation (1), except epy = 3 kv	---	---	tj	---	0.005	µs
	Quality conformance inspection, part 3								
---	Life test	9	Group C; t = 96 hours "on" and "1" hour "off"; t = 600 hours	---	---	---	---	---	---
---	Life - test end points:								
3246	Operation (1) and (1A)	-		---	---	egy	---	150	v
3247	DC anode voltage for conduction	-	egy = 150 v	---	---	Ebb	---	1,000	v
3261	Time jitter	-	egy = 150 v	---	---	tj	---	0.005	µs
(A) 1031	Sweep - frequency vibration	4, 8	0 to 2,000 Hz	---	---	---	---	---	---
1041	Shock	4, 16	1000; no voltages applied	---	---	---	---	---	---
---	Sweep - frequency vibration and shock end points:								
3246	Operation (1)	-		---	---	egy	---	150	v
3247	DC anode voltage for conduction	-		---	---	Ebb	---	300	Vdc
3261	Time jitter	-		---	---	tj	---	0.005	µs
(A) 3246	Operation at elevated ambient temperature	4, 9, 15	TA = 150°C; t = 5 hours	---	---	egy	---	150	v

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NOTES:

1. Instantaneous starting is permissible. The maximum permissible instantaneously applied epy is 8.0 kv and shall not be attained in less than 0.04 seconds. The epy may then be raised to full rating.

2. In pulse operation, the peak inverse voltage, exclusive of a spike of 0.05 µs maximum duration, shall not exceed 2.0 kv during the first 25 µs following the anode pulse.

3. The driver pulse, measured at tube socket with thyratron grid disconnected shall have the following characteristics: 175 volts minimum; 600 volts maximum; tr = 0.35 µs maximum; tp = 2.0 µs minimum, impedance of drive circuit Zs = 1,500-ohms maximum. At -55°C, 250 V minimum shall be required.

Ⓐ 4. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

5. The tube is capable of operation of over 50,000 pps within the limitations of the Pb and Ib current ratings. With a saturable reactor, Pb equal to 4 x 10^5 is permissible for certain applications.

6. It may be desirable to employ forced-air cooling under conditions of high Pb number operations. A cooling airblast of 5 cfm may be directed into the anode cap.

7. Appreciably less jitter than 0.005 µs can be realized if the anode voltage is 3 kv or more, grid drive amplitude is near the maximum and grid drive impedance is near minimum.

8. There shall be no pronounced resonance in the range from 0 to 2,000 Hz.

9. The circuit constants shall be chosen under resonant charging conditions so that: epy = 14 kv; ib = 150 a minimum; dik/dt = 1,500 a/µs minimum; tp = 1.0 ± 10 percent; prr = 1,000 minimum; Rgl = 30,000 ohms. Grid pulse characteristics shall be: tr = 0.35 µs minimum; tp = 2.0 µs maximum, and driver impedance = 500-ohms minimum.

10. The tube shall operate satisfactorily on push-button starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 8 kv minimum within 0.03 seconds. (The filter in the rectifier shall be designed so that the epy reaches at least 4 kv within 0.015 second.)

11. This test shall be conducted within 60 seconds of the operation (1) test.

12. The tube shall operate continuously for 10 minutes.

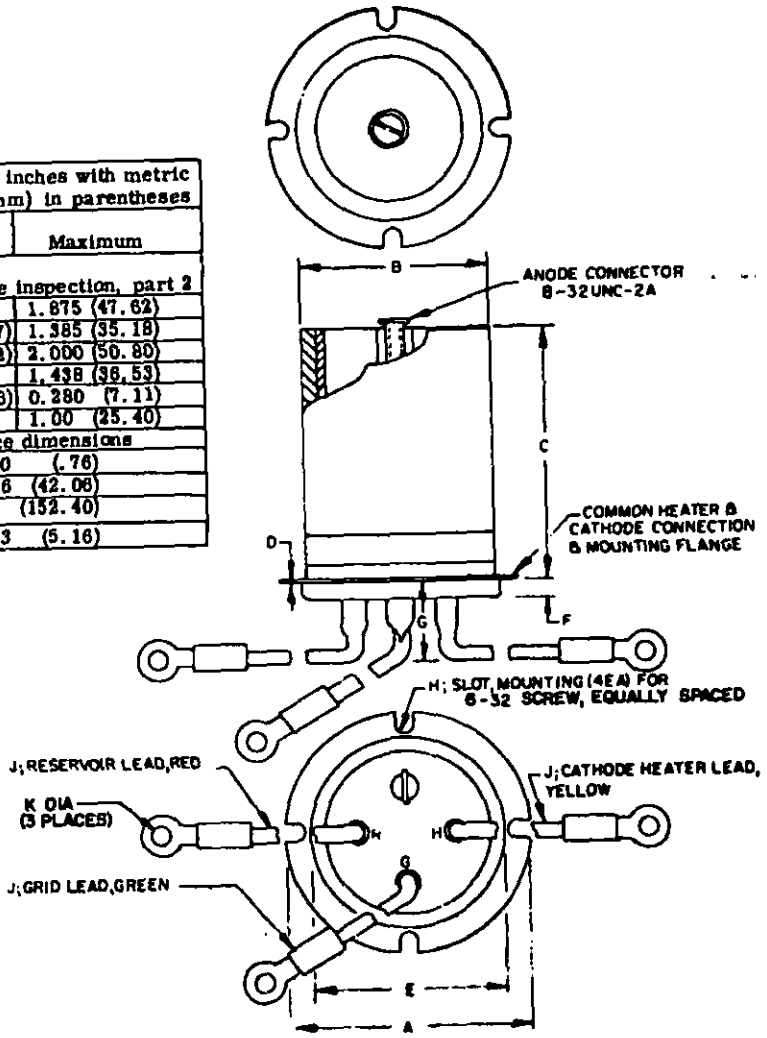
13. This test shall be performed simultaneously with the operation (1) test. An anode delay time measurement shall be made at the end of 2 and 10 minutes of the operation (1) test. The change in anode delay time (with respect to the 2-minute reading) shall not exceed the specified value at any time during this test.

14. The tube shall be tested by applying a peak forward anode voltage not to exceed that specified in the test conditions for the time jitter test immediately after the cathode warmup period (tk). The variation in firing time (tj) shall not be greater than the amount specified after 60 seconds of operation.

15. This test shall be conducted for a total of 6 consecutive hours with no more than 3 kickouts and with no evidence of detrimental anode heating. The tube shall be started with Eres 107-1/2 percent Vac and operate at this value for 4 hours. At the start of the fifth hour, and while the tube is still operating, the filament voltage shall be lowered to Eres = 92-1/2 percent Vac and remain there for the final hour of operation.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		1.875 (47.62)
B	1.365 (34.67)	1.385 (35.18)
C	1.875 (47.62)	2.000 (50.80)
E		1.438 (36.53)
F	0.156 (3.96)	0.280 (7.11)
G		1.00 (25.40)
Reference dimensions		
D	0.030 (.76)	
H	1.656 (42.06)	
J	6.00 (152.40)	
K	.203 (5.16)	



Ⓐ FIGURE 1. Outline drawing of electron tube type 7782.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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